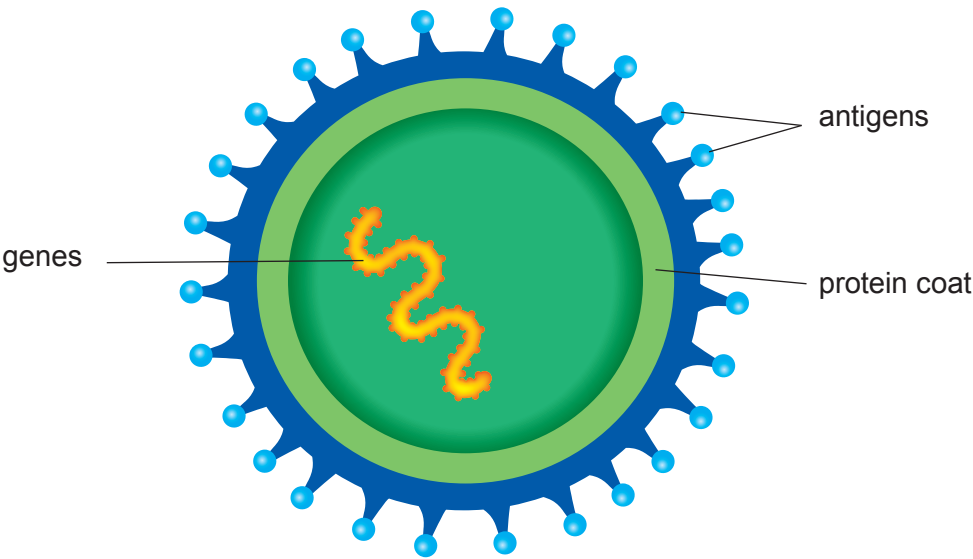


3. (a) The diagram shows a pathogen that causes an infectious disease.



(i) State the type of pathogen shown above. [1]

.....

(ii) Sudden changes to a gene cause changes to the antigens on the coat of the pathogen.

State the term for a sudden change to a gene. [1]

.....

(b) State **one** way by which pathogens are spread between people. [1]

.....

(c) The following statements (**1 – 6**) describe how the body responds when it is invaded by pathogens. The statements are not in the correct order.

- 1 the pathogens have foreign antigens
- 2 pathogens enter the blood
- 3 they respond by releasing antibodies to the antigens
- 4 the antibodies attach to the pathogen
- 5 the antigens are recognised as foreign by lymphocytes
- 6 the attached antibodies help phagocytes destroy the pathogen

Complete the sequence to put the statements in the correct order. [3]

..... **1** **3**

6

